SMART DUSTBIN PROJECT

Review 2 Report – 35% Design and Documentation Stage

1. PROJECT OVERVIEW

The project is a Smart Waste Bin Monitoring and AI-Based Overflow Detection System designed to address overflowing public waste bins and inefficient waste collection. A field study was conducted in Goundanur, where overflowing public bins, lack of proper wet/dry waste segregation, and limited daily collection were observed.

The proposed system uses an ESP32 microcontroller and HC-SR04 ultrasonic sensor to estimate the fill level of a dustbin. When the fill level reaches an initial threshold of 80%, the system generates an alert indicating that the bin requires collection. A future AI module will detect visible waste overflow using images and object-detection techniques.

At the Review 2 stage, approximately 35% of the project has been completed, covering field study, problem identification, objectives, system architecture, hardware selection, sensing methodology, alert logic, algorithm design, prototype planning, and technical documentation.

2. PROBLEM STATEMENT

Public waste bins can become full or overflow before the next scheduled collection. Manual monitoring of every bin is difficult, while fixed collection schedules do not always reflect the actual condition of each bin. Overflowing waste can create unhygienic surroundings.

The proposed solution is an automated monitoring system that identifies the fill condition of a bin and provides an early alert before significant overflow occurs.

3. OBJECTIVES

The main objectives are:

- Automatically monitor waste-bin fill level.
- Use an HC-SR04 ultrasonic sensor to measure the distance to the waste surface.
- Interface the sensor with an ESP32.
- Convert measured distance into an estimated fill percentage.
- Generate an alert when the fill level reaches approximately 80%.
- Reduce continuous manual inspection.
- Support condition-based waste collection.
- Develop a future AI module for visible overflow detection.
- Provide a foundation for multiple smart bins.

4. SYSTEM ARCHITECTURE

The basic architecture is:

```
Waste Bin → HC-SR04 → ESP32 → Distance Measurement → Fill-Level Calculation → 80% Threshold → Alert
```

The ultrasonic sensor is positioned near the top of the bin. It sends an ultrasonic pulse toward the waste surface and measures the returning echo. The ESP32 calculates the distance and converts it into a fill percentage using the known effective height of the bin.

If the calculated fill level is 80% or greater, the system identifies a critical condition and generates an alert.

5. HARDWARE COMPONENTS

The planned prototype consists of:

- ESP32 – main controller for reading and processing sensor data.

- HC-SR04 ultrasonic sensor – measures distance to the waste surface.
- Dustbin – physical environment for testing.
- Breadboard and jumper wires – used for prototype assembly.
- Alert/output mechanism – indicates the critical fill condition.

6. FILL-LEVEL CALCULATION

The fill percentage can be calculated using:

```
Fill Percentage = ((Bin Height - Measured Distance) / Bin Height) × 100
```

For example, if the effective bin height is 50 cm and the measured distance is 10 cm:

```
Fill Percentage = ((50 - 10) / 50) × 100
Fill Percentage = 80%
```

The actual values will be calibrated according to the dimensions and sensor position of the physical dustbin.

7. ALERT LOGIC

The initial threshold is 80%.

- Below 80% – Normal condition.
- At or above 80% – Critical condition and alert.

The threshold can be adjusted after physical testing. The purpose is to provide an early warning so collection can take place before the bin reaches complete capacity.

8. ALGORITHM / PSEUDOCODE

```
START
```

Initialize ESP32 and HC-SR04.

Set the known bin height.

```
Repeat:
Send ultrasonic pulse.
Read echo time.
Calculate distance.
Calculate fill percentage.
If fill percentage >= 80%:
Generate alert.
Else:
Continue normal monitoring.
Display distance and fill percentage.
```

Repeat.

```
STOP
```

9. CURRENT PROJECT STATUS

The following work has been completed for the 35% stage:

- Field study and real-world problem identification.
- Problem statement and objectives.
- Proposed system architecture.
- ESP32 and HC-SR04 component selection.

- Fill-level measurement concept.
- Fill percentage calculation method.
- 80% threshold alert logic.
- Algorithm and pseudocode.
- Prototype planning.
- AI overflow-detection concept.
- Technical and project documentation.

The documentation clearly separates the proposed design from features that are still under implementation.

10. AI-BASED OVERFLOW DETECTION

The project also proposes an AI-based image detection module to identify visible overflow.

The planned workflow is:

```
Camera/Image → Image Processing → Object/Overflow Detection → Classification → Alert
```

An object-detection approach such as YOLO may be used. Possible categories include normal bin, nearly full bin, overflowing bin, and waste outside the bin.

This AI component is currently part of the planned development and will be implemented and tested in the next stage.

11. TESTING PLAN

The prototype will be tested under different fill conditions:

- Empty bin – low fill percentage and normal status.
- Partially filled bin – intermediate fill percentage.
- Approximately 80% filled bin – threshold alert.
- Overflow condition – critical/overflow indication.

Serial-monitor output will be captured to demonstrate actual distance and fill-level readings.

Expected output:

```
Distance: 35 cm
Fill Level: 30%
Status: NORMAL
Distance: 20 cm
Fill Level: 60%
Status: NORMAL
Distance: 10 cm
Fill Level: 80%
Status: ALERT - BIN 80% FULL
```

Actual readings will depend on the physical bin dimensions and sensor position.

12. AREAS FOR IMPROVEMENT AND NEXT STEPS

The immediate next steps are:

1. Develop and upload a working Arduino/ESP32 .ino sketch to read the HC-SR04, calculate distance and fill percentage, and trigger the 80% alert.

2. Create the actual circuit/schematic showing the HC-SR04 and ESP32 pin connections.

3. Assemble the circuit on a breadboard and mount the sensor on an actual dustbin.

4. Capture photographs of the physical prototype and wiring.

5. Capture serial-monitor evidence showing distance, fill percentage, and alert status.

6. Begin AI overflow detection by preparing a labelled sample image dataset and/or an initial YOLO/object-detection implementation.

13. FUTURE SYSTEM

The final system is intended to combine sensor-based monitoring with AI-based visual detection.

```
Ultrasonic Sensor → ESP32 → Fill-Level Monitoring → Alert
```

```
Camera → AI Detection → Visible Overflow Detection → Alert
```

In the future, multiple smart bins could send their status to a centralized monitoring platform. This could help sanitation authorities identify bins requiring collection and improve collection efficiency.

14. CONCLUSION

The Review 2 stage has established a clear foundation for the Smart Waste Bin Monitoring and AI-Based Overflow Detection System. The project has progressed from real-world field observation to a defined technical solution using an ESP32 and HC-SR04 ultrasonic sensor.

The system architecture, objectives, hardware selection, fill-level calculation, 80% threshold logic, algorithm, pseudocode, prototype plan, and AI integration concept have been completed as part of the 35% project stage.

The major remaining work is practical implementation: developing the ESP32 code, building and documenting the physical circuit, collecting live sensor readings, and starting the AI-based overflow detection module. These steps will provide implementation evidence and move the project from design and documentation toward a tested working prototype.

CURRENT STATUS: 35% DESIGN AND DOCUMENTATION STAGE COMPLETED. HARDWARE IMPLEMENTATION, LIVE SENSOR TESTING, AND AI OVERFLOW DETECTION ARE THE NEXT DEVELOPMENT PRIORITIES.